

Installing plugins

When Abot boots, it starts every plugin listed in your *plugins.json* file.

Open up *plugins.json* in your text editor of choice, and add to the *Dependencies*, so it looks like what's shown below:

```
{
  "Dependencies": {
    "github.com/itsabot/plugin_onboard": "",
    "github.com/itsabot/plugin_weather": ""
  }
}
```

Then, in a terminal, run the following:

```
$ abot plugin install
```

```
Fetching 2 plugins...
Fetching dependencies...
Installing plugins...
Success!
```

```
$ abot server
```

And from another terminal, run the following:

```
$ abot c +918889992221
```

```
> What's the weather like in Mumbai?
It's 74 and sunny in Mumbai.
```

Creating your first Abot plugin

Abot provides all the tools you need to quickly and easily write these plugins through its shared library. To do so, you should know the Go programming language.

Let's take a look at a 'Hello World' plugin, which will introduce you to the plugin API. Let's download the plugin by adding it to our *plugins.json* file and installing it as follows:

```
$ tail $GOPATH/src/github.com/itsabot/abot/plugins.json
{
  ...
  "Dependencies": {
    "github.com/itsabot/plugin_onboard": "",
    "github.com/itsabot/plugin_hello": ""
  }
}

$ abot plugin install
Fetching 2 plugins...
Installing plugins...
Success!
```

Now, let's take a look at the contents of the plugin. Pay close attention to the inline comments, which explain each piece of the plugin API as it is introduced.

```
// Package hello responds to "Say something" with "Hello World".

package hello

import (
    "log"

    "github.com/itsabot/abot/shared/datatypes"
    "github.com/itsabot/abot/shared/nlp"
    "github.com/itsabot/abot/shared/plugin"
)

var p *dt.Plugin
var sm *dt.StateMachine

func init() {
    // When Abot receives a message, it'll route the message to the correct
    // package. Doing that requires a trigger, which tells Abot to send the
    // response to this package when Commands include "say" and Objects
    // include "something", "hello", etc. Case should always be lowercase,
    // and the words will be stemmed automatically, so there's no need to
    // include variations like "cat" and "cats".
    trigger := &nlp.StructuredInput{
        Commands: []string{"say"},
        Objects:  []string{"something", "hello", "hi"},
    }

    // Tell Abot how we'll respond to first messages and follow up messages
    // from a user in a conversation.
    fns := &dt.PluginFns{Run: Run, FollowUp: FollowUp}

    // Create the package, setting it up to communicate with Abot through
    // the functions we specified.
    var err error
    p, err = plugin.New("github.com/itsabot/plugin_hello", trigger, fns)
    if err != nil {
        log.Fatalln("building", err)
    }

    // Abot includes a state machine designed to have conversations. This
```